Resume Annotator

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Motivation:

- Building a resume for a Job description can be a tedious task.

- In this project we are addressing this critical issue and make it easier for Users, In Understanding mistakes and changes can be made to their resume.

We:

1. Created a Interface Using Streamlit where user has to upload resume and description of the job user is applying

2. Analyse the Resume and Job description

3. Provided ATS score of the resume w.r.t the Description with suggestions of analysis

4. Finally highlight the changes to be made in resume helping user decide where and what changes are to be made.

5. Mail resume to the user extracted from Resume

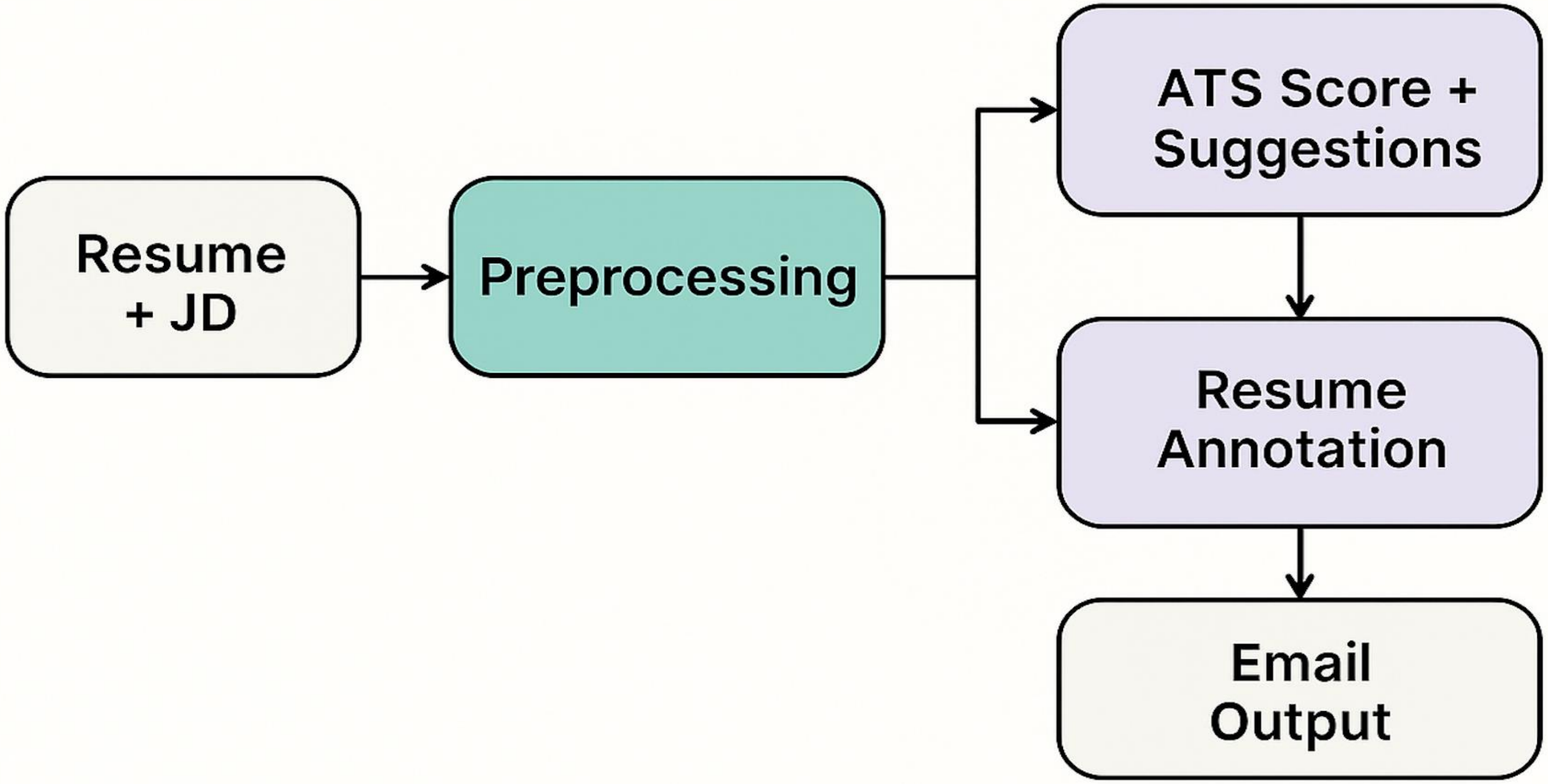
Sources

- Data collected from User Resume and Job Description.
- Custom Data on Rules, Restrictions, Guidelines, Examples for LLM
- Kaggle Resume Dataset

Packages and Models

- Streamlit, Mistral AI, GPT-4, BERT-xtremedistil-l6-h256-uncased, re, spacy, collections ,Counter, transformers, sentence_transformers and More available at Requirements

Project Workflow



Preprocessing:

Here in the initial stages we did Text extraction from Job description and Resume

ATS score:

ATS Score is the sum of aggregated components such as

- Keyword Match Score
- Experience Alignment Score
- Competency & Skill Match Score
- Formatting & ATS Compliance Score
- Sentence Structure & Impact Analysis Score
- Job Description Mirroring Score

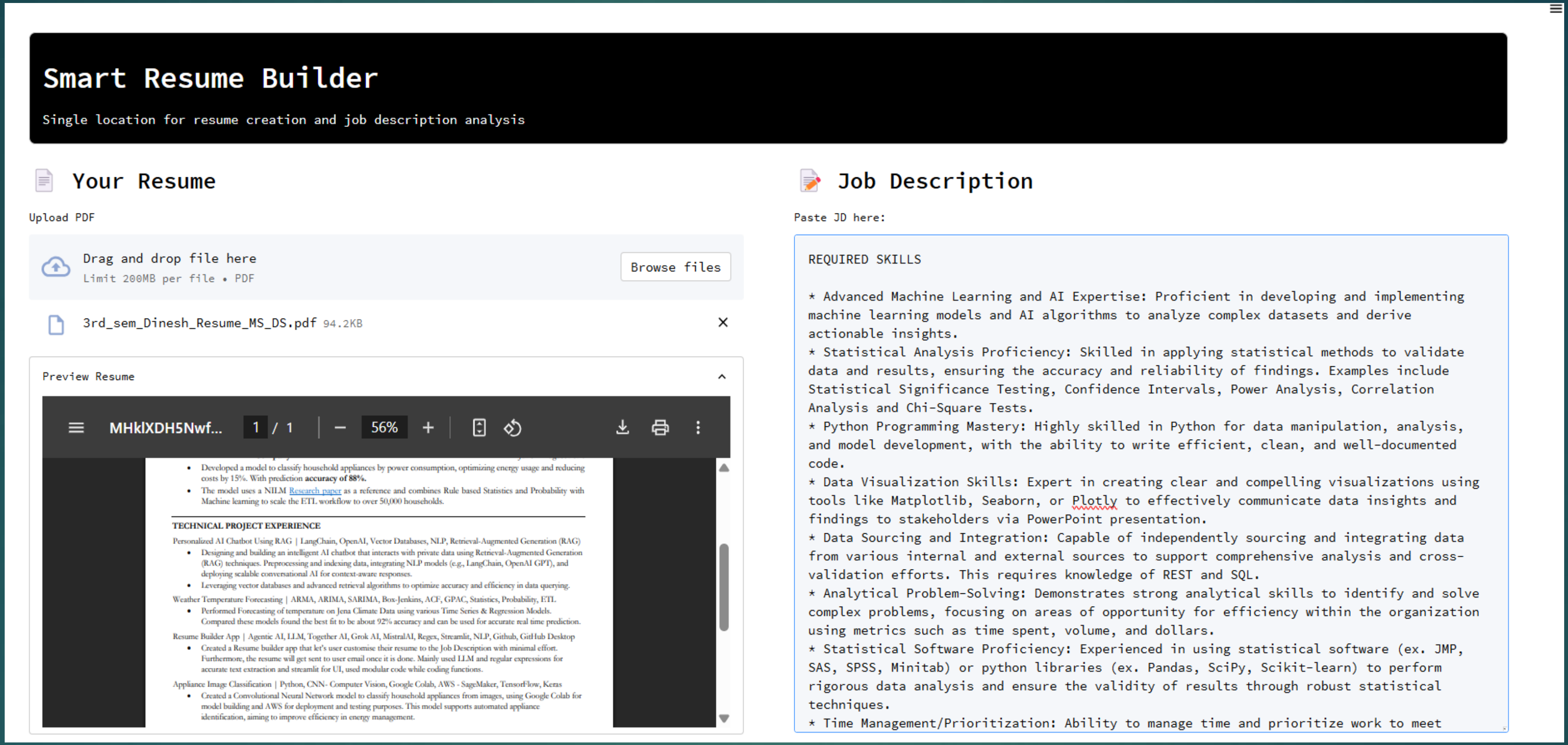
Resume Annotations:

- Red – Action verbs
- Green- metrics
- blue - Impact
- Hard Skills – orange
- Soft – yellow

Email notification:

We used Smtblib, Email to mail user his Annotated resume.

User Interface:



User uploads his Resume and relevant Job description in this landing page and clicks Analyse which does the preprocessing of the data.

UI Page-2

Page-2 has ATS analytics, Resume which gets Annotated and Side bar that has ATS metrics

Deploy

app

Resume Builder

Match Rate

75%

Upload & rescan

⚡ Power Edit

Searchability – 2 issues to fix

Hard Skills – 1 issue to fix

Soft Skills – 2 issues to fix

Recruiter Tips – 1 issue to fix

Formatting – 2 issues to fix

Smart Resume Builder

Single location for resume creation and job description analysis

ATS Analytics | Job Description | Resume

📌 Job Description Focus Analysis

Label Frequency

- **ACTION:** 64 mentions
- **HARD:** 48 mentions
- **SOFT:** 28 mentions
- **IMPACT:** 19 mentions
- **METRIC:** 1 mentions

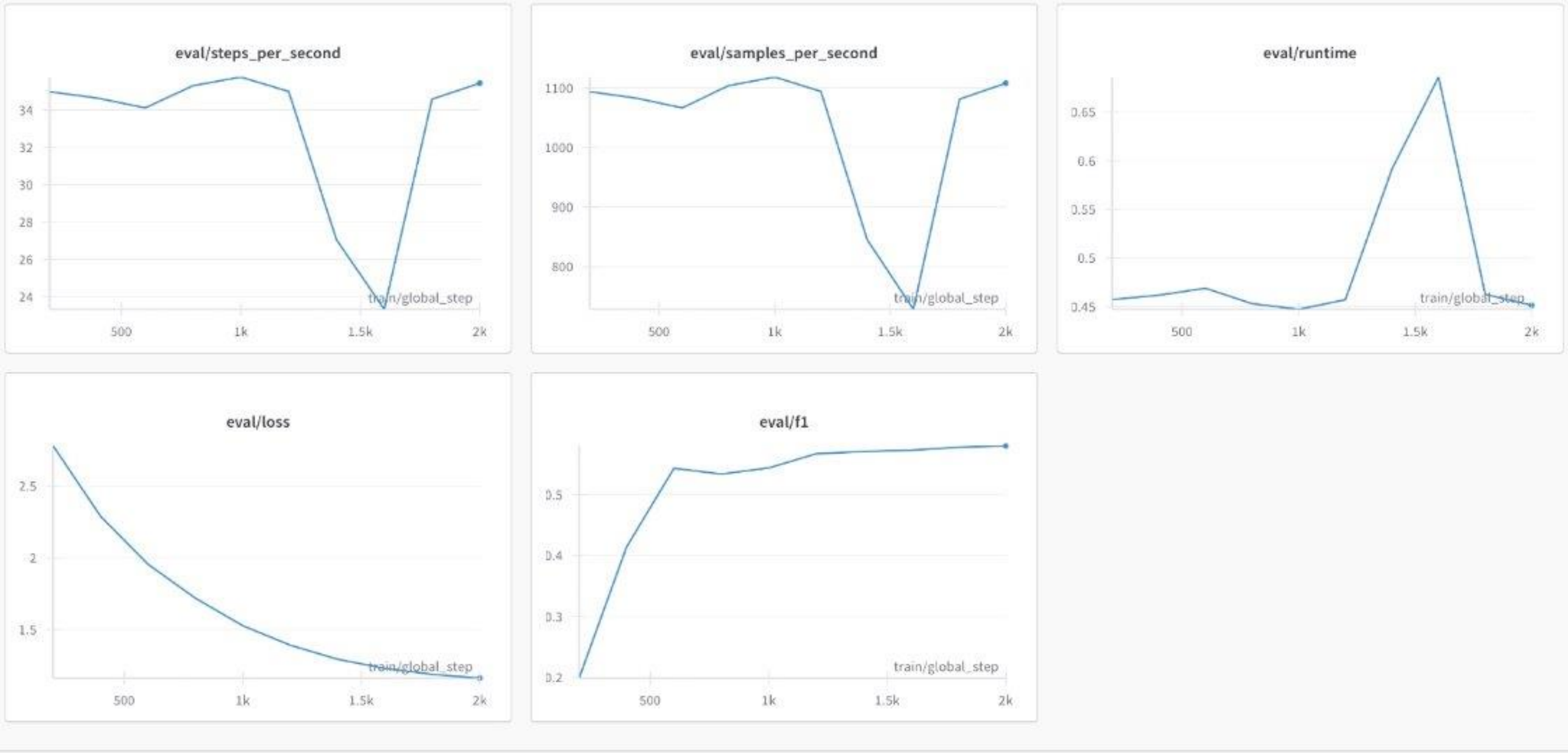
💡 Insight

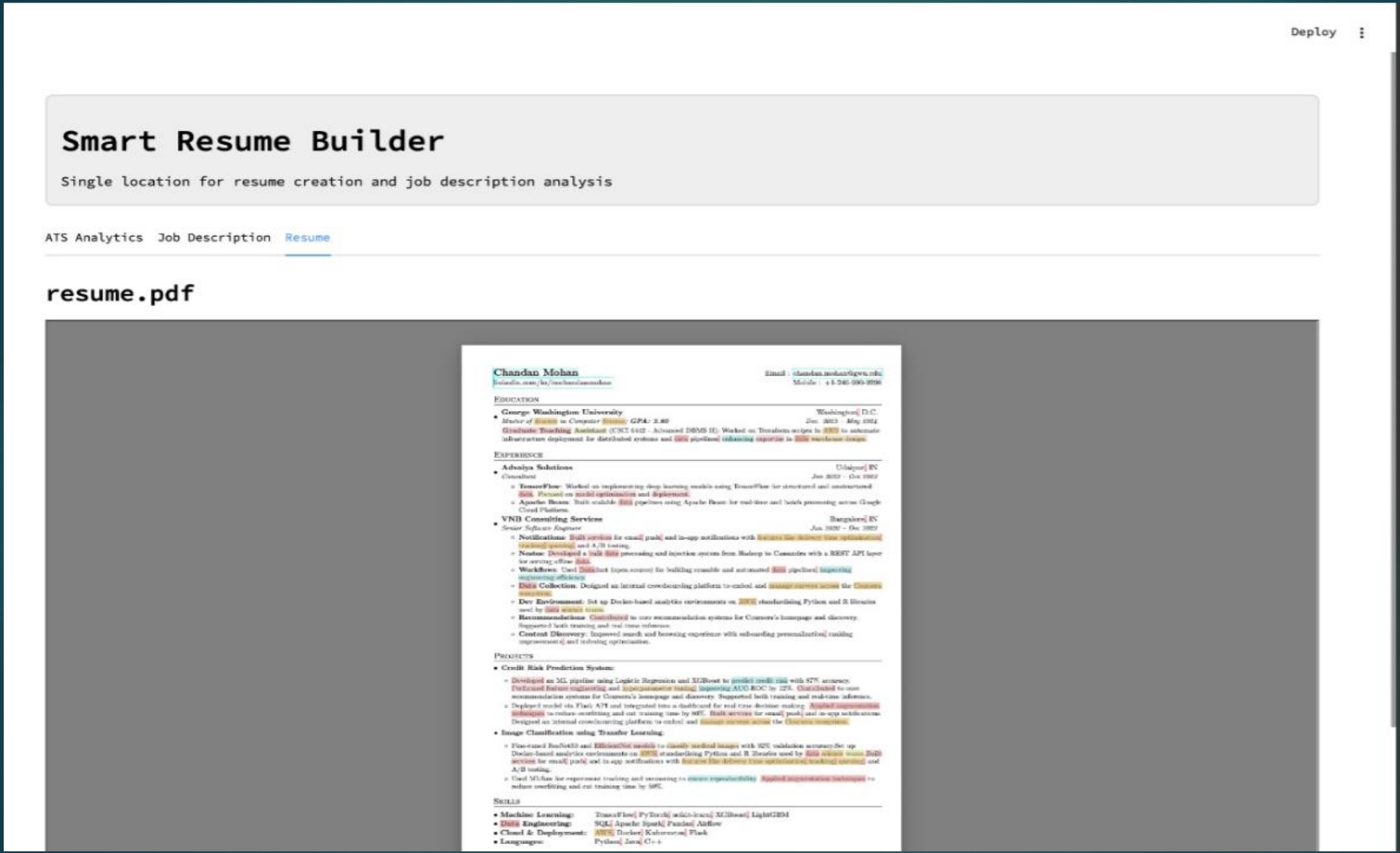
This job description emphasizes ACTION, HARD, SOFT.

👉 Make sure your resume uses strong action verbs to describe contributions, and highlights technical and platform-specific skills, and shows collaboration and communication abilities.

Bert Evaluation Metrics Used in Annotation:

"activation": "gelu",
"architectures":
  "DistilBertForTokenClassification",
"attention_dropout": 0.1,
"dim": 256,
"hidden_dim": 512,
"hidden_dropout": 0.1,
"initializer_range": 0.02,
"max_position_embeddings": 512,
"n_heads": 4,
"n_layers": 6,
"vocab_size": 30522,




Deploy
Smart Resume Builder
Single location for resume creation and job description analysis
ATS Analytics
Job Description
Resume
resume.pdf

Results

Resume Analysis optimized for ATS

Sent to user's email

User Resume Annotated for Editing

References:

- Streamlit. (n.d.). *Streamlit documentation*. https://streamlit.io
- Scikit-learn Developers. (n.d.). *TF-IDF vectorizer — scikit-learn 1.4.1 documentation*. https://scikit-learn.org
- Bird, S., Klein, E., & Loper, E. (n.d.). *Natural Language Toolkit (NLTK)*. https://www.nltk.org
- Together AI. (n.d.). *Together AI documentation*. https://docs.together.ai

Thank you.